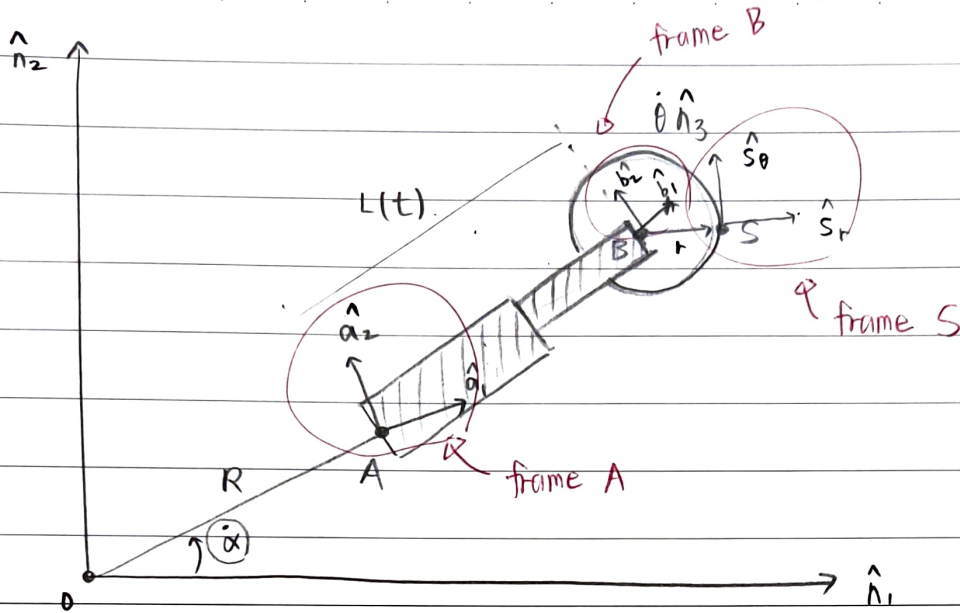


< kinematics final assignment 6/21 >

①



Part I)

$$\vec{r} = R\hat{a}_1 + L(t)\hat{b}_1 + r\hat{s}_1 = \vec{r}_{S/N}$$

Angular Velocity	{	$W_{A/N} = \dot{\alpha}\hat{n}_3$	frame	{	$N: \{\hat{n}_1, \hat{n}_2, \hat{n}_3\}$
		$W_{B/N} = \dot{\beta}\hat{n}_3$			$A: \{\hat{a}_1, \hat{a}_2, \hat{n}_3\}$
		$W_{S/B} = \dot{\theta}\hat{n}_3$			$B: \{\hat{b}_1, \hat{b}_2, \hat{n}_3\}$
		$W_{S/N} = W_{S/B} + W_{B/N}$			$S: \{\hat{s}_1, \hat{s}_2, \hat{n}_3\}$
		$= (\dot{\theta} + \dot{\beta})\hat{n}_3$			

$$\begin{aligned} \dot{\vec{r}}_{S/N} &= \frac{d}{dt}(\vec{r}) = \frac{d}{dt}(R\hat{a}_1 + L(t)\hat{b}_1 + r\hat{s}_1) \quad \text{using transport theorem.} \\ &= \left\{ \frac{d}{dt}(R\hat{a}_1) + W_{A/N} \times (R\hat{a}_1) \right\} + \left\{ \frac{d}{dt}(L(t)\hat{b}_1) + W_{B/N} \times (L(t)\hat{b}_1) \right\} \\ &\quad + \left\{ \frac{d}{dt}(r\hat{s}_1) + W_{S/N} \times (r\hat{s}_1) \right\} \\ &= \{W_{A/N} \times (R\hat{a}_1)\} + \{L(t)\dot{\hat{b}}_1 + W_{B/N} \times (L(t)\hat{b}_1)\} + \{W_{S/N} \times (r\hat{s}_1)\} \end{aligned}$$

$$\vec{F}_{S/N} = (\ddot{\alpha} \hat{n}_3) \times (R \hat{a}_1) + \dot{L}(t) \hat{b}_1 + (\dot{\beta} \hat{n}_3) \times (\dot{L}(t) \hat{b}_1) + (\dot{\theta} + \dot{\beta}) \hat{n}_3 \times (r \hat{s}_r)$$

$$= R \ddot{\alpha} \hat{a}_2 + \dot{L}(t) \hat{b}_1 + \dot{\beta} \dot{L}(t) \hat{b}_2 + r(\dot{\theta} + \dot{\beta}) \hat{s}_\theta$$

Part 2) To get acceleration, do transport theorem about velocity ($\vec{F}_{S/N}$) of point S

$$\ddot{\vec{r}}_{S/N} = \frac{d}{dt}(\vec{F}_{S/N}) = \frac{d}{dt} \left(R \ddot{\alpha} \hat{a}_2 + \dot{L}(t) \hat{b}_1 + \dot{\beta} \dot{L}(t) \hat{b}_2 + r(\dot{\theta} + \dot{\beta}) \hat{s}_\theta \right)$$

$$= \left\{ \frac{d}{dt} (R \ddot{\alpha} \hat{a}_2) + \omega_{A/N} \times (R \ddot{\alpha} \hat{a}_2) \right\} + \left\{ \frac{d}{dt} (\dot{L}(t) \hat{b}_1) + \omega_{B/N} \times (\dot{L}(t) \hat{b}_1) \right\} \\ + \left\{ \frac{d}{dt} (\dot{\beta} \dot{L}(t) \hat{b}_2) + \omega_{B/N} \times (\dot{\beta} \dot{L}(t) \hat{b}_2) \right\} + \left\{ \frac{d}{dt} (r(\dot{\theta} + \dot{\beta}) \hat{s}_\theta) \right. \\ \left. + \omega_{S/N} \times (r(\dot{\theta} + \dot{\beta}) \hat{s}_\theta) \right\}$$

$$= \left\{ R \ddot{\alpha} \hat{a}_2 + \ddot{\alpha} \hat{n}_3 \times (R \ddot{\alpha} \hat{a}_2) \right\} + \left\{ \ddot{L}(t) \hat{b}_1 + (\dot{\beta} \hat{n}_3) \times (\dot{L}(t) \hat{b}_1) \right\}$$

$$+ \left\{ (\ddot{\beta} \dot{L}(t) + \dot{\beta} \ddot{L}(t)) \hat{b}_2 + (\dot{\beta} \hat{n}_3) \times (\dot{\beta} \dot{L}(t) \hat{b}_2) \right\}$$

$$+ \left\{ r \ddot{\beta} \hat{s}_\theta + (\dot{\theta} + \dot{\beta}) \hat{n}_3 \times (r(\dot{\theta} + \dot{\beta}) \hat{s}_\theta) \right\}$$

$$= R \ddot{\alpha} \hat{a}_2 - R \ddot{\alpha}^2 \hat{a}_1 + \ddot{L}(t) \hat{b}_1 + \dot{\beta} \ddot{L}(t) \hat{b}_2 + \left\{ \ddot{\beta} \dot{L}(t) + \dot{\beta} \ddot{L}(t) \right\} \hat{b}_2$$

$$- \dot{\beta}^2 \dot{L}(t) \hat{b}_1 + r \ddot{\beta} \hat{s}_\theta - r(\dot{\theta} + \dot{\beta})^2 \hat{s}_r$$

$$= R \ddot{\alpha} \hat{a}_2 - R \ddot{\alpha}^2 \hat{a}_1 + \left\{ \ddot{L}(t) - \dot{\beta}^2 \dot{L}(t) \right\} \hat{b}_1 + \left\{ \ddot{\beta} \dot{L}(t) + 2 \dot{\beta} \ddot{L}(t) \right\} \hat{b}_2 \\ + r \ddot{\beta} \hat{s}_\theta - r(\dot{\theta} + \dot{\beta})^2 \hat{s}_r$$

Part 3)

$$\mathbf{r}_{A/S} = -r\hat{\mathbf{s}}_r - L(t)\hat{\mathbf{b}}_1$$

$${}^S \frac{d}{dt}(\mathbf{r}_{A/S}) = {}^S \frac{d}{dt}(-r\hat{\mathbf{s}}_r - L(t)\hat{\mathbf{b}}_1)$$

$$= {}^S \frac{d}{dt}(-L(t)\hat{\mathbf{b}}_1)$$

$$- \omega_{S/B} = -\dot{\theta}\hat{\mathbf{n}}_3$$

$$= {}^B \frac{d}{dt}(-L(t)\hat{\mathbf{b}}_1) + \omega_{B/S} \times (-L(t)\hat{\mathbf{b}}_1)$$

$$= -\dot{L}(t)\hat{\mathbf{b}}_1 + \{\dot{\theta}\hat{\mathbf{n}}_3 \times L(t)\hat{\mathbf{b}}_1\}$$

$$= -\dot{L}(t)\hat{\mathbf{b}}_1 + \dot{\theta}L(t)\hat{\mathbf{b}}_2$$

②

$$[B(\sigma)] = (1-\sigma^2)[I_{3 \times 3}] + 2[\tilde{\sigma}] + 2\sigma\sigma^T$$

$$\begin{aligned} [B(\sigma)]^T &= \left((1-\sigma^2)[I_{3 \times 3}] \right)^T + 2[\tilde{\sigma}]^T + (2\sigma\sigma^T)^T \\ &= (1-\sigma^2)[I_{3 \times 3}] + 2[\tilde{\sigma}]^T + 2\sigma\sigma^T \end{aligned}$$

$$\begin{aligned} [B][B]^T &= \left\{ (1-\sigma^2)[I_{3 \times 3}] + 2[\tilde{\sigma}] + 2\sigma\sigma^T \right\} \left\{ (1-\sigma^2)[I_{3 \times 3}] + 2[\tilde{\sigma}]^T + 2\sigma\sigma^T \right\} \\ &= (1-\sigma^2)^2 [I_{3 \times 3}] + 2(1-\sigma^2)[\tilde{\sigma}]^T + (1-\sigma^2)2\sigma\sigma^T \\ &\quad + 2(1-\sigma^2)[\tilde{\sigma}] + 4[\tilde{\sigma}][\tilde{\sigma}]^T + 4[\tilde{\sigma}]\sigma\sigma^T \\ &\quad + 2(1-\sigma^2)\sigma\sigma^T + 4\sigma\sigma^T[\tilde{\sigma}]^T + 4\sigma\sigma^T\sigma\sigma^T \end{aligned}$$

since $[\tilde{\sigma}]$: skew-symmetric $[\tilde{\sigma}]^T = -[\tilde{\sigma}]$

$$\begin{aligned} [B][B]^T &= (1-\sigma^2)^2 [I_{3 \times 3}] - 2(1-\sigma^2)[\tilde{\sigma}] + (1-\sigma^2)2\sigma\sigma^T \\ &\quad + 2(1-\sigma^2)[\tilde{\sigma}] - 4[\tilde{\sigma}]^2 + 4[\tilde{\sigma}]\sigma\sigma^T \\ &\quad + 2(1-\sigma^2)\sigma\sigma^T - 4\sigma\sigma^T[\tilde{\sigma}] + 4\sigma\sigma^T\sigma\sigma^T \end{aligned}$$

since $[\tilde{\sigma}]\sigma = 0$, $\sigma^T[\tilde{\sigma}] = 0$, $[\tilde{\sigma}]^2 = \sigma\sigma^T - \sigma^2[I_{3 \times 3}]$

$$\begin{aligned} [B][B]^T &= (1-\sigma^2)^2 [I_{3 \times 3}] + 4(1-\sigma^2)\sigma\sigma^T + 4\sigma^2\sigma\sigma^T - 4\sigma\sigma^T + 4\sigma^2[I_{3 \times 3}] \\ &= (1+\sigma^2)^2 [I_{3 \times 3}] \end{aligned}$$

$$[B][B]^T = (1+\sigma^2)^2 I, \text{ which means } [B] \left(\frac{[B]^T}{(1+\sigma^2)^2} \right) = I, [B]^{-1} = \frac{[B]^T}{(1+\sigma^2)^2}$$

$$\text{Therefore } [B]^{-1} = \frac{[B]^T}{(1+\sigma^2)^2} = \frac{1}{(1+\sigma^2)^2} \left\{ (1-\sigma^2)[I_{3 \times 3}] - 2[\tilde{\sigma}] + 2\sigma\sigma^T \right\}$$

< Answers >

$$1 - \text{part 1} : R\ddot{\alpha}\hat{a}_z + \dot{L}(t)\hat{b}_1 + \dot{\beta}L(t)\hat{b}_2 + r(\dot{\theta} + \dot{\beta})\hat{S}_\theta$$

$$1 - \text{part 2} : R\ddot{\alpha}\hat{a}_z - R\dot{\alpha}^2\hat{a}_1 + \left\{ \ddot{L}(t) - \dot{\beta}^2L(t) \right\} \hat{b}_1 \\ + \left\{ \dot{\beta}L(t) + 2\dot{\beta}\dot{L}(t) \right\} \hat{b}_2 + r\ddot{\beta}\hat{S}_\theta - r(\dot{\theta} + \dot{\beta})^2\hat{S}_r$$

$$1 - \text{part 3} : -\dot{L}(t)\hat{b}_1 + \dot{\theta}L(t)\hat{b}_2$$

$$2 : [B]^{-1} = \frac{1}{(1+\sigma^2)^2} \left\{ (1-\sigma^2)[I_{3 \times 3}] - 2[\tilde{\sigma}] + 2\sigma\sigma^T \right\}$$